Multirate Factor Analysis Models for Fault Detection in Multirate Processes

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***Abstract*—Generally, the measurements of modern industries are collected from different sources, which indicates that the traditional multivariate statistical process monitoring methods cannot be directly used in the multirate systems if one aims to utilize the complete multirate measurements. Hence, a set of multirate factor analysis models is developed for process modeling and fault detection purpose in the multirate processes. In the proposed model, the cross correlations are described and bounded by the common factors and the model parameters are calibrated using the expectation–maximum algorithms. Also, the proposed models are further discussed both from theoretical and geometric perspective. Finally, the proposed fault detection methods are tested by a simulated Tennessee–Eastman process and a real R2S anaerobic reactor unit in the wastewater treatment process.**

***Index Terms*—EM algorithm, fault detection, multirate factor analysis, multirate process.**

I. Introduction

IN RECENT two decades, the data-based process monitoring has been well studied and widely applied in both academic and industrial areas [1]–[5]. Especially, the multivariate statistical process monitoring (MSPM) methods are quite popular techniques since the improvement of the machining learning, deep learning, and the widespread applications of distribution control systems [6]–[9]. Jiang and Yin have provided a detailed review on the recent advances in the data-based prognosis and diagnosis approaches oriented to the key-performance indicators and developed a MATLAB toolbox DB-KIT [10]. Ge has reviewed different kinds of basic probabilistic latent variable models for process monitoring purpose [11]. Jiang *et al.* have investigated the recent researches in process monitoring, fault diagnosis, and control approaches based on data-driven technologies in the industrial cyber-physical systems [12].

However, most traditional MSPM methods are built under the assumption that all the observations are sampled and saved in an uniform rate [13], [14]. In real industries, the observations of an operation plant or even a crucial unit are actually collected from different sources. Generally, the variables which reveal the process operating status are sampled by the hardware sensors and recorded per seconds or minutes. Besides, some key quality indicators need the laboratory analysis due to the equipment limitations or actual demands. These variables are usually sampled with a large range of the sampling rates (from 30 min to several weeks). Moreover, some special detection signals are sampled in an even faster or lower rate. For instance, the electric current is sampled per millisecond. To sum up, it is desirable to develop an MSPM method using the multirate sampled instead of the uniformly sampled observations.

For multirate process modeling and monitoring purpose, the traditional methods are to make the pretreatment for the multisampling-rate measurements so that they are transferred to the uniformly sampled forms. Usually, the main techniques include downsampling and upsampling. For downsampling methods, one aims to find out the lowest sampling rate so that all the measurements can be uniformly recorded in this rate. In contrary, the upsampling methods try to use the measurements with highest sampling rates through estimating the corresponding low-sampled variables. Among them, Tham *et al.* used an accurate and robust artificial neural network for the prediction of the infrequent measurements of quality variables [15]. Li *et al.* studied identifying a fast single-rate model using the multirate input–output data, where a lifted state-space model is used to consider the causality constraint [16]. Besides, Tipping and Bishop proposed probabilistic principal component analysis model and they pointed out the probabilistic model is suitable for handling the missing values and it can be used as the estimator in the upsampling method [17]. However, no matter downsampling or upsampling methods, the original cross-correlation structures of the multisampling-rate measurements will be screwed more or less.

Hence, it is more proper to directly use the multisampling-rate measurements without downsampling or upsampling. For this purpose, the dual-sampling-rate systems are first studied. One choice is to use the data fusion techniques. Lu *et al.* developed two models which take both the high-sampling-rate process variables and the low-sampling-rate quality variables [18]. Wu and

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Luo used a data fusion technology and the multirate Kalman filter approach is adopted for the improvement of the accuracy and reliability estimations [19]. Another kinds of methods are the so-called semisupervised learning. A semisupervised probabilistic latent variable regression model is proposed for process monitoring in both continuous and batch process [20]. Bao *et al.* has adopted cotraining algorithms for partial least squares to incorporate both labeled and unlabeled data samples for soft sensors [21]. Ge and Song developed a semisupervised Bayesian method using a small portion of output and enough input samples. Also, the dimensionality of the latent spaces can be properly determined [22].

Inspired by the semisupervised methods in the dual-sampling-rate processes, a multirate model is expected to be subsequently implemented to incorporate all the measurements in the process with multiple sampling rates. In our previous work, two kinds of pioneer works has been proposed. The first one is multirate PLS [23], which separated the complete covariance of the multisampling-rate measurements into several independent covariances. However, this model is only valid under the assumption that the distributions of the missing values of fast-sampling-rate data are the same as the collected one and it is too restrict in real applications. Besides, taking advantage of the probabilistic models, a multirate probability PCA (MRPCA) model was proposed for fault detection purpose in a triple-sampling-rate process [24]. In the probabilistic framework, the common latent variables of different sampling rates can be estimated in nature instead of calculating the separated covariance in MRPCA. Furthermore, there are still two aspects which need to be improved. At first, a more common model structure, model training procedure, and the corresponding fault detection schemes should be derived for any multisampling-rate processes. Next, the assumption that the noises distribution in PPCA is isotropic is too rigorous.

For solving these problems, we have utilized the traditional factor analysis (FA) model and designed a set of multirate models based on the single-sampling-rate FA. Hence, the proposed method is named as multi-rate factor analysis (MR-FA) models. Moreover, FA has designed with a clear objective to find out the certain unobservable factors and it will also reveal a clear interpretations on the underlying construction of measurements [25], [26]. Hence, it is more proper to be extended to a multirate form. In the proposed model, the common factors are shared by all the measurements with different sampling rates. And, it can be found that both the intracorrelations between the individual sampling rate and the intercorrelations between different sampling rates are considered to determine the common factors. Hence, the proposed MR-FA models are totally different from several independent FA models and they are able to extract the inherent features in a multisampling-rate process under a more realistic assumption.

The remainder of this paper is structured as follows. At first, the traditional PPCA and FA model is briefly reviewed in Section II. Next, the proposed MR-FA models are introduced and explained in details, which also follows the model training steps, the discussions on the model properties, and the corresponding fault detection scenario. In Section IV, a simulated Tennessee–Eastman (TE) process and a real R2S anaerobic reactor unit in the wastewater treatment process (WWTP) are demonstrated to validate the efficiency of the proposed method. Finally, some conclusions are given.

II. Preliminaries

In this section, a short revisit on PPCA and FA model is given in succession. After that, the differences between these two models are briefly discussed.

A. Probability Principal Component Analysis

In an industrial process which contains the single-sampling-rate variables, the traditional dimension reduction method like PCA or PPCA is often used. Among them, the measurements $\mathbf{x}$ can be derived under the probability framework using PPCA and it is given as

$$\mathbf{x} = \mathbf{\Phi}\mathbf{t} + \mathbf{e} \tag{1}$$

where $\mathbf{t}$ are the latent variables and they are assumed to follow a prior Gaussian distribution with zero mean and variance $\mathbf{I}$, where $\mathbf{I}$ is the identity matrix. In the probability framework, the post distribution of the latent variables and the measurements can be estimated given the loading matrix $\mathbf{\Phi}$. Also, the process noises are assumed to follow Gaussian distributions with zero mean and variance $\sigma^2\mathbf{I}$. As a kind of generative model, PPCA is based on the mapping from the latent spaces to the data spaces, which is different from the traditional projection direction in PCA [27]. In PPCA, however, we have assumed that the variances of the process noises are isotropic. In the static process, it is reasonable that the noises of different process variables are independent. However, their magnitudes should be various rather than isotropic. Hence, it is desirable to relax the restriction on this limitations.

B. Factor Analysis Model

FA model is another kind of linear Gaussian latent variable model which is closely related to PPCA [27]. It describes the correlated variables in terms of a low-dimensional unobserved variables called factors. It is supposed that a set of measurements $\mathbf{x}$ are collected, FA model is given as

$$\mathbf{x} = \mathbf{\Psi}\mathbf{t} + \varepsilon \tag{2}$$

where $\mathbf{t}$ are called factors and $\mathbf{\Psi}$ are the factor loadings. Also, ε are unobserved stochastic error terms with zero mean and finite variance, which is given as $\varepsilon \sim \mathrm{N}(\mathbf{0},\mathbf{\Omega})$ and $\mathbf{\Omega}$ is a diagonal matrix.

Even though both FA and PCA (or PPCA) are dimension reduction techniques, the approaches taken to do are diverse. The model assumptions, interpretabilities, and solutions between FA and PCA are all different. At first, PCA accounts for a mathematical transformation to the original data with no assumptions about the form of the covariance matrix. FA can be treated as a form of latent variable density model and it is assumed that the distributions of the measurement noises are taken to have a

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diagonal rather than an isotropic covariance [28]–[31]. Besides, it is suggested that FA is designed with a clear objective to find out the certain unobservable factors, while PCA or PPCA fail to do so [25]. Recently, Brown has pointed that FA has considered the random error that is inherent in measurements [26]. Hence, he recommends using FA when the theoretical relationships between the variables are clear, whereas PCA is suggested to be used if one wants to explore the main patterns in the data. In PCA, the components do not represent underlying constructions and it aims to determine several principal components that represent the data without losing much information. Whereas, FA not only indicates an accurate dimension reduction results, but also supplies a labeled and readily interpretations on the underlying construction of data. Finally, the model solutions of PCA and FA are also different. PCA is performed by the eigenvector decomposition, while the factor loadings of FA are usually obtained using an iterative manner such as the EM algorithm. Links between FA and PCA are also investigated. It is pointed that the maximum of the log-likelihood function in FA occurs when the eigenvectors are the principal eigenvectors [17]. It is also noticed that the traditional PCA, PPCA, and FA are all taken using the uniformly sampling data. If we take into account the practical use in complex industrials, it is desirable to extend the traditional method to its multisampling-rate form. Based on the discussion on PCA and FA, we have developed the multirate FA models in next section due to its several advantages over PCA.

III. Multirate-Factor-Analysis-Based Fault Detection

In this section, the MR-FA models are first defined, which is followed by the detailed model parameter estimation procedures. After that, some discussions and geometric interpretations of the proposed MR-FA model will be introduced. Finally, the MR-FA-based fault detection scenario is given.

A. Multirate Factor Analysis Models

In an industrial process with N kinds of sampling rates, which is noted as $\mathbf{X}_1 \in R^{K_1 \times M_1}, \mathbf{X}_2 \in R^{K_2 \times M_2}, \ldots, \mathbf{X}_N \in R^{K_N \times M_N}$, we can design the MR-FA models as

$$\mathbf{x}_n(k) = \mathbf{\Psi}_n \mathbf{t}(k) + \boldsymbol{\xi}_n(k), n = 1, 2, \ldots, N \tag{3}$$

in which $\mathbf{x}_n(k)$ is one of the observations at the sample time k in the matrix $\mathbf{X}_n$ and it is readily to obtain that $\mathbf{X}_n = \{\mathbf{x}_n(1), \mathbf{x}_n(2), \ldots, \mathbf{x}_n(k), \ldots, \mathbf{x}_n(K_n)\}$. It can be seen that both of the number of samples and variables may be various for different sampling rates. Since N kinds of sampling rates are collected, there are also N corresponding factor loadings, which are written as $\mathbf{\Psi}_n \in R^{M_n \times D}, n = 1, 2, \ldots, N$. Also, the measurements noises are $\boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_N$. The prior distributions of measurement noises follow Gaussian distributions as $\boldsymbol{\xi}_n \sim N(\mathbf{0}, \mathbf{\Omega}_n)$ and $\mathbf{\Omega}_n$ is the diagonal matrix. Different from several separated FA model, the factors $\mathbf{t}(k)$ in the proposed MR-FA model are shared with all the sampling rates at the sampling time k. Hence, it can be seen that the factor matrix is $\mathbf{T} = \{\mathbf{t}(1), \mathbf{t}(2), \ldots \mathbf{t}(K)\} \in R^{K \times D}$, in which the total sampling time interval for the training model can be noted as K. And, at least one kind of sampling-rate variables must be recorded at each sampling interval k, which indicates that $K \geq \max\{K_1, K_2, \ldots, K_N\}$. It is noticed that K can be more than the maximum sample number of one sampling rate if the sampling interval for different sampling rate is not common divisor. To make sure that the factors are uncorrelated, we also make several assumptions as $\mathbf{t}(k)$ and $\boldsymbol{\xi}_n$ are independent, $E(\mathbf{t}(k)) = \mathbf{0}$ and $\mathrm{Cov}(\mathbf{t}(k)) = \mathbf{I}$, and $\mathbf{I}$ is the unit matrix. Given the measurements with different sampling rates and the factor loadings, the model inferences of the proposed MR-FA can be taken using the probability theory and Bayes' theorem. The model parameters $\{\mathbf{\Psi}_n, \mathbf{\Omega}_n\}, n = 1, 2, \ldots, N$ of MR-FA is determined by the maximum likelihood. Since there is no longer a closed-form maximum likelihood solution in FA-related models, the EM algorithm is used in this paper, which is derived in detail in the next subsection.

B. Model Parameter Estimation Using EM

The model parameters $\mathbf{\Psi}_n$ and $\mathbf{\Omega}_n$ in MR-FA are estimated using an iterative manner called EM algorithm. In the E-step, the present model parameters are fixed for estimating the posterior distributions of factors and they are further updated using the maximum likelihood in the M-step.

In the proposed MR-FA model, the posterior distributions of factors at the sampling time k are estimated as

$$\hat{\mathbf{t}}(k) = \mathbf{\Sigma}(k)^{-1} \left(\sum_{n=1}^{N} \phi_{kn} \mathbf{\Psi}_n^T \mathbf{\Omega}_n^{-1} \mathbf{x}_n(k) \right) \tag{4}$$

$$\mathbf{\Sigma}(k) = \sum_{n=1}^{N} \phi_{kn} \mathbf{\Psi}_n^T \mathbf{\Omega}_n^{-1} \mathbf{\Psi}_n + \mathbf{I} \tag{5}$$

$$E(\mathbf{t}(k)\mathbf{t}^T(k) \,|\mathbf{X}^o(k)) = \mathbf{\Sigma}(k)^{-1} + \hat{\mathbf{t}}(k)\hat{\mathbf{t}}(k)^T \tag{6}$$

in which $\hat{\mathbf{t}}(k) = E(\mathbf{t}(k) \,|\mathbf{X}^o(k))$ is the expectation of factors based on the collected measurements at the sampling time k and they are noted as $\mathbf{X}^o(k)$. Also, another crucial index is the sampling indicator ϕ_{kn}, which is used to reveal the sampling state of nth sampling rate at the current sampling time. We have defined that ϕ_{kn} equals to 1 if $\mathbf{x}_n(k)$ are sampled, whereas it will be 0 when $\mathbf{x}_n(k)$ are not sampled. Using $\mathbf{X}^o(k)$ and ϕ_{kn}, the collection state at the sampling time k is clearly expressed. For instance, if $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_{N-1}$ have been collected at the sampling time k, it can be obtained that $\mathbf{X}^o(k) = \left[\mathbf{x}_1(k)\ \mathbf{x}_2(k)\ \mathbf{x}_{N-1}(k)\right] \in R^{(M_1+M_2+M_{N-1})\times 1}$, $\phi_{k1} = \phi_{k2} = \phi_{kN-1} = 1$, and $\phi_{k3} = \cdots = \phi_{kN-2} = \phi_{kN} = 0$.

Based on the estimated results in the E-step, the complete likelihood can be calculated as

$$L = \sum_{k=1}^{K} \ln p(\mathbf{X}^o(k), \hat{\mathbf{t}}(k) \big| \mathbf{\Psi}_1, \mathbf{\Psi}_2, \ldots, \mathbf{\Psi}_n, \mathbf{\Omega}_1, \mathbf{\Omega}_2, \ldots, \mathbf{\Omega}_n). \tag{7}$$

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By maximizing the likelihood, the model parameters are updated in the M-step as

$$\hat{\mathbf{\Psi}}_n = \left[\sum_{k=1}^{K} \phi_{kn} \mathbf{x}_n(k) \hat{\mathbf{t}}^T(k)\right] \left[\sum_{k=1}^{K} \phi_{kn} E(\mathbf{t}(k)\mathbf{t}^T(k) \,|\mathbf{X}^o(k))\right]^{-1} \tag{8}$$

$$\hat{\mathbf{\Omega}}_n = \text{diag}\left\{\mathbf{S}_n - \frac{1}{K_n}\sum_{k=1}^{K} \phi_{kn} \hat{\mathbf{\Psi}}_n \hat{\mathbf{t}}(k) \mathbf{x}_n^T(k)\right\} \tag{9}$$

where diag(.) is the calculator to make all the nondiagonal elements of a matrix to zero. $\mathbf{S}_n$ is the covariance matrix of the *n*th sampling rate data, which is given as

$$\mathbf{S}_n = \frac{1}{K_n}\sum_{k=1}^{K_n} \mathbf{x}_n(k)\mathbf{x}_n^T(k). \tag{10}$$

It is noticed that the total time interval used in (8) and (9) are all *K* even though the overall sampling time is various from K_1 to K_N for different sampling rates. In such way, a uniform formula can be obtained for different sampling-rate data instead of several separated equations. And it can be achieved by using the sampling indicators ϕ_{kn}. For the *n*th sampling rate data, ϕ_{kn} would be zero for the sampling interval in which these variables are not collected. Hence, the total calculated sample number of $\mathbf{x}_n$ are still K_n in (8) and (9). After several iterations between the E-step and the M-step, the likelihood will be converged to a local maximum value and the estimated model parameters can be obtained. However, the efficiency of the proposed model will be affected if the different sampling rates are mutually coprime. In this case, only when the sampling interval is the common multiple of these coprime numbers, the factors will be determined by all these sampling-rate measurements and the correlations between them can be described.

C. Discussions and Geometric Interpretations of MR-FA

In this subsection, some properties of the proposed MR-FA models are further discussed by several remarks, which are followed by their geometric interpretation.

Remark 1: From the training procedures, it can be seen that the proposed MR-FA is a special time-variant model. The factors at the sampling time *k* is estimated using (4) and the used corresponding model structures like factor loadings and the variance $\mathbf{\Sigma}(k)$ changes with time interval since they are determined by how many kinds of sampling rates are collected at the current time. Hence, it indicates that the model structures are time variant and they are chosen from $\{\mathbf{\Psi}_n, \mathbf{\Omega}_n\}, n = 1, 2, \ldots, N$, which is achieved by the sampling indicators ϕ_{kn}.

Remark 2: The proposed MR-FA can be treated as a more general form of FA model. Actually, when $N = 1$, MR-FA will reduce to the traditional FA. Furthermore, MR-FA is reduced to the supervised or semisupervised FA, in which $N = 2$ and $K_1 = K_2$ for the supervised case and $N = 2$ and $K_1 \neq K_2$ for the semisupervised one.

Remark 3: It is quite different between several separated FA models and the proposed MR-FA. If one sets up the individual FA models for each sampling rates, these models are totally independent and the covariance between different sampling rates are not taken into consideration. However, the common factors are shared by different sampling rates in MR-FA, which build the relationship between them. Moreover, each factor loading is determined by all the collected data instead of the corresponding sampling rate data themselves.

The Remark 3 will be further explained from both the mathematical and geometric perspectives. First, if we substitute (4)–(6) into (8)–(9), it can be obtained that

$$\begin{aligned}\hat{\mathbf{\Psi}}_n &= \left[\sum_{k=1}^{K}\sum_{j=1}^{N} \phi_{kn} \mathbf{x}_n(k) \mathbf{x}_j{}^T(k) \mathbf{\Omega}_j^{-1} \mathbf{\Psi}_j \mathbf{\Sigma}^T(k)^{-1}\right] \\ &\quad \times \left[\sum_{k=1}^{K} \phi_{kn} \left(\mathbf{\Sigma}(k)^{-1} + \mathbf{\Sigma}(k)^{-1}\mathbf{A}(k)\mathbf{\Sigma}(k)^{-1}\right)\right]^{-1} \\ &= \left[\sum_{k=1}^{K}\sum_{j=1}^{N} \phi_{kn} \mathbf{x}_n(k) \mathbf{x}_j{}^T(k) \mathbf{\Omega}_j^{-1} \mathbf{\Psi}_j \mathbf{\Sigma}^T(k)^{-1}\right] \\ &\quad \times \left[\sum_{k=1}^{K} \phi_{kn} \left(\mathbf{I} + \mathbf{\Sigma}(k)^{-1}\mathbf{A}(k)\right) \mathbf{\Sigma}(k)^{-1}\right]^{-1}\end{aligned} \tag{11}$$

$$\mathbf{A}(k) = \sum_{n=1}^{N}\sum_{j=1}^{N} \phi_{kn} \mathbf{\Psi}_n^T \mathbf{\Omega}_n^{-1} \mathbf{x}_n(k) \mathbf{x}_j{}^T(k) \mathbf{\Omega}_j^{-1} \mathbf{\Psi}_j \tag{12}$$

$$\hat{\mathbf{\Omega}}_n = \text{diag}\left\{\mathbf{S}_n - \frac{1}{K_n}\sum_{k=1}^{K} \phi_{kn} \hat{\mathbf{\Psi}}_n \mathbf{\Sigma}(k)^{-1} \mathbf{B}^T(k)\right\} \tag{13}$$

$$\mathbf{B}(k) = \sum_{j=1}^{N} \mathbf{x}_n(k) \mathbf{x}_j{}^T(k) \mathbf{\Omega}_j^{-1} \mathbf{\Psi}_j. \tag{14}$$

Indeed, the model parameters can also be estimated using (11)–(14), which indicates that the updated values of $\hat{\mathbf{\Psi}}_n$ and $\hat{\mathbf{\Omega}}_n$ are determined by its own old values and the collected data. And it is noticed that the factor loading for the *n*th sampling rate $\mathbf{\Psi}_n$ are estimated not only using the measurements of the *n*th sampling rate, whereas all kinds of correlated measurements of other sampling rates are involved. From (12), it can be seen that the correlated information between $\mathbf{x}_n(k)$ and all the other sampling-rate measurements at the sampling time *k* is calculated when ϕ_{kn} equals to 1, which is also true for $\hat{\mathbf{\Omega}}_n$. Hence, the proposed MR-FA has built the correlated relationship between different sampling rates even though their factor loadings are estimated separately. In Fig. 1, a sketch map of MR-FA with three sampling rates are shown for instance to reveal a visualization of the model constructions. It can be seen that the factor loadings $\mathbf{\Psi}_1$, $\mathbf{\Psi}_2$, and $\mathbf{\Psi}_3$ are determined by all the correlated measurements. In such manner, an uniformed multisampling-rate model has been built instead of several individual models.

Next, the geometric interpretations of the proposed MR-FA are introduced. In the single FA model, the measurements and noises are the vectors in the *M*-dimensional Euclidean space and the factors will define a *D*-dimensional hyperplane, on which the measurements are orthogonally projected [32]. For easy visualization, we can take a three-dimensional (3-D) vector for

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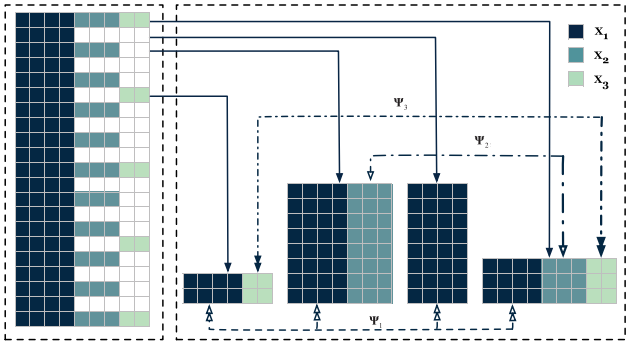

Fig. 1. Sketch map of MR-FA with three sampling rates.

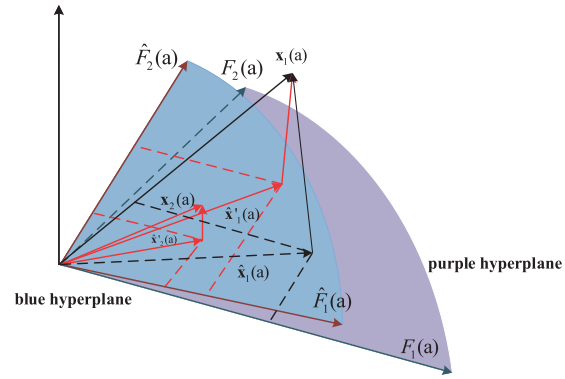

Fig. 2. Geometric interpretation of MR-FA.

instance. For the normalized vector $\mathbf{x}(a)$, which is projected onto a plane defined by two orthogonal vectors F_1 and F_2. It can be obtained that $\mathbf{x}(a) = \hat{\mathbf{x}}(a) + \boldsymbol{\xi}(a)$ since the projected vector $\hat{\mathbf{x}}(a)$ and the noises $\xi(a)$ are orthogonal. If another vector $\mathbf{x}(b)$ is collected, the correlation matrix for $\mathbf{x}(a)$ and $\mathbf{x}(b)$ would be $r_{ab} = \mathbf{x}(a)\mathbf{x}(\mathrm{b})$ since these measurements or vectors have been normalized. From the geometrical perspective, the correlation matrix can be viewed as the cosine of the angle between these two vectors. Also, we can define the projected correlation matrix $\hat{r}_{ab}$ as $\hat{r}_{ab} = \hat{\mathbf{x}}(a)\hat{\mathbf{x}}(\mathrm{b})$.

Actually, the objective of FA model is to define a best-fit hyperplane in which the projected correlation matrix is able to represent the original data relationship as far as possible. Hence, the mean square error in the off-diagonal components should be minimized when the fitting hyperplane is obtained, which is given as

$$\boldsymbol{\xi}^2 = \sum_{a,b\neq a} (r_{ab} - \hat{r}_{ab})^2. \tag{15}$$

Also, it is readily to obtain that $r_{ab} - \hat{r}_{ab} = \boldsymbol{\xi}(a)\boldsymbol{\xi}(\mathrm{b})$, which indicates that the goal of FA can be achieved in terms of the minimization of the covariance of the errors, which are given in the term on the right of (15). Since the error covariance has been assumed to be a diagonal matrix in FA, a best-fit model will just be accomplished by minimizing the off-diagonal components of the error covariance in the mean square sense.

For the MR-FA model, however, the goal becomes to find out a best-fit hyperplane using the multirate data. Even though the dimension for different sampling rate data may be various, they are still be projected onto the same hyperplane due to the common factors shared by all of them. Specifically, for the fitting hyperplane, the mean square errors in the off-diagonal components of the proposed MR-FA model are given as

$$\begin{aligned}\boldsymbol{\xi}^2_{\text{all}} &= \sum_{i,j\neq i} (r_{ij,1} - \hat{r}_{ij,1})^2 \cdots \sum_{i,j\neq i} (r_{ij,N} - \hat{r}_{ij,N})^2 \\ &= \sum_{i,j\neq i} (\boldsymbol{\xi}_{i,1}\boldsymbol{\xi}_{j,1})^2 \sum_{i,j\neq i} (\boldsymbol{\xi}_{i,2}\boldsymbol{\xi}_{j,2})^2 \cdots \sum_{i,j\neq i} (\boldsymbol{\xi}_{i,N}\boldsymbol{\xi}_{j,N})^2\end{aligned} \tag{16}$$

where $r_{ij,n}$ is the correlation matrix for two vectors of nth sampling rate and $\hat{r}_{ij,n}$ is the corresponding projected correlation matrix. $\boldsymbol{\xi}_{i,n}$ and $\boldsymbol{\xi}_{j,n}$ are their covariance of the errors.

Hence, the total objective of MR-FA is to search the global minimization of the covariance of the errors for all the sampling rate data instead of the local minimization of the covariance of the errors for each individual sampling rate data. At this point, Remark 3 is further explained on the difference between MR-FA and several individual FA models. From the geometric perspective, two 2-D factors are determined for instance, which is shown in Fig. 2. It can be seen that the original purple hyperplane is determined by two factors $F_1(a)$ and $F_2(a)$ given the single-sampling-rate vectors $\mathbf{x}_1(a)$ using FA model. The corresponding projected vector is given as $\hat{\mathbf{x}}_1(a)$. When another kind of sampling rate data $\mathbf{x}_2(a)$ has been incorporated, MR-FA is used to determine another blue hyperplane, which is given in terms of two new orthogonal factors $\hat{F}_1(a)$ and $\hat{F}_2(a)$. The corresponding projected vectors are written as $\hat{\mathbf{x}}'_1(a)$ and $\hat{\mathbf{x}}'_2(a)$. And the new best-fit hyperplane becomes to find out the minimization of the covariance of the errors for these two kinds of measurements. Hence, due to the use of different sampling rate data, the hyperplane determined by factors will rotate to search a more proper solution and it has been clearly revealed in the geometric viewpoint.

D. MR-FA-Based Fault Detection

In this subsection, the multiple fault detection scenario based on the proposed MR-FA is given using two commonly used statistics T^2 and SPE. Usually, T^2 is adopted to detect the variations of the subspace determined by factors, while the residual subspace is monitored using SPE statistics. Specially, we have built several SPE statistics for each sampling rate data to reveal the variations of the respective residual subspace.

The T^2 statistic is constructed based on the common factors. For the new collected data $\mathbf{x}_n(k_{\text{new}})$, the factors and the corresponding T^2 statistic at the current sampling time is given as

$$\hat{\mathbf{t}}(k_{\text{new}}) = \boldsymbol{\Sigma}(k_{\text{new}})^{-1} \left(\sum_{n=1}^{N} \phi_{k_{\text{new}}n} \boldsymbol{\Psi}_n^T \boldsymbol{\Omega}_n^{-1} \mathbf{x}_n(k_{\text{new}}) \right) \tag{17}$$

$$T^2_{k_{\text{new}}} = \hat{\mathbf{t}}^T(k_{\text{new}}) \boldsymbol{\Sigma}(k_{\text{new}})^{-1} \hat{\mathbf{t}}(k_{\text{new}}) \tag{18}$$

where $\boldsymbol{\Sigma}(k_{\text{new}}) = \sum_{n=1}^{N} \phi_{k_{\text{new}}n} \boldsymbol{\Psi}_n^T \boldsymbol{\Omega}_n^{-1} \boldsymbol{\Psi}_n + \mathbf{I}$. Similarly, the SPE statistics are built using the residuals of each sampling rate

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TABLE I
FALSE ALARMS AND MISSED DETECTION RATES OF 21 FAULTS IN TE PROCESS

Fault	MR-FA				MRPCA				FA_1		FA_2	
	T^2	SPE_1	SPE_2	SPE_3	T^2	SPE_1	SPE_2	SPE_3	T^2	SPE	T^2	SPE
IDV(0)	0.031	0.013	0.000	0.000	0.025	0.006	0.000	0.000	0.000	0.125	0.009	0.011
IDV(1)	0.001	0.009	0.006	**0.005**	0.006	0.003	0.013	0.450	0.025	0.025	0.013	0.006
IDV(2)	0.013	0.040	0.019	0.050	0.025	0.015	0.019	0.050	0.025	0.025	0.030	0.018
IDV(3)	**0.868**	0.973	0.931	0.825	0.935	0.963	0.969	0.850	0.975	0.700	0.973	0.983
IDV(4)	0.924	0.990	0.956	**0.775**	0.955	0.980	0.981	0.975	1.000	0.850	0.989	0.990
IDV(5)	**0.674**	0.908	**0.794**	**0.625**	0.724	0.778	0.925	0.750	0.775	0.625	0.743	0.763
IDV(6)	0.000	0.000	0.006	0.025	0.000	0.000	0.013	0.050	0.025	0.000	0.008	0.000
IDV(7)	**0.518**	0.774	0.681	0.600	0.568	0.716	0.763	0.525	0.575	0.475	0.595	0.629
IDV(8)	0.016	0.184	0.038	**0.175**	0.030	0.029	0.069	0.250	0.025	0.025	0.041	0.023
IDV(9)	**0.886**	0.976	0.938	0.975	0.941	0.970	0.975	0.950	0.975	0.900	0.974	0.979
IDV(10)	**0.106**	0.773	**0.375**	**0.275**	0.541	0.713	0.969	0.750	0.475	0.450	0.518	0.623
IDV(11)	**0.529**	0.559	**0.888**	0.975	0.810	0.549	0.975	0.900	0.950	0.525	0.960	0.601
IDV(12)	0.005	0.283	0.138	0.175	0.011	0.039	0.225	0.125	0.050	0.025	0.013	0.014
IDV(13)	0.050	0.093	0.050	0.100	0.059	0.050	0.050	0.075	0.075	0.050	0.050	0.061
IDV(14)	**0.010**	0.000	**0.738**	0.750	0.106	0.000	0.900	0.425	0.325	0.050	0.258	0.000
IDV(15)	**0.844**	0.984	0.938	0.850	0.928	0.970	0.994	0.850	0.950	0.725	0.961	0.978
IDV(16)	**0.355**	0.961	**0.825**	**0.725**	0.680	0.846	0.944	0.825	0.600	0.575	0.660	0.848
IDV(17)	**0.050**	0.049	**0.388**	0.450	0.175	0.044	0.575	0.375	0.325	0.050	0.215	0.068
IDV(18)	0.086	0.109	0.094	0.125	0.108	0.100	0.106	0.100	0.125	0.075	0.111	0.098
IDV(19)	**0.626**	**0.666**	0.894	0.925	0.815	0.826	0.969	0.850	0.950	0.650	0.961	0.810
IDV(20)	**0.190**	0.538	**0.456**	**0.575**	0.640	0.590	0.869	0.725	0.450	0.350	0.446	0.645
IDV(21)	**0.368**	0.576	**0.650**	0.725	0.528	0.531	0.819	0.800	0.450	0.550	0.389	0.641

data, respectively, which is given as

$$e_n(k_{\text{new}}) = \mathbf{x}_n(k_{\text{new}}) - \mathbf{\Psi}_n \hat{\mathbf{t}}(k_{\text{new}}) \tag{19}$$

$$\text{SPE}_n = e_n^T e_n \tag{20}$$

where SPE_n represents the statistic for the nth sampling rate. The control limits of T^2 and SPE_n statistics can be estimated by the F-distribution and the chi-square distribution, respectively, which can be referred in [33].

IV. CASE STUDY

In this section, two examples are demonstrated to validate the performance of the proposed MR-FA-based fault detection schemes. The first one is a simulated TE process, which has been widely used as a benchmark for evaluating the process monitoring performance among different algorithms. Next, a real R2S anaerobic reactor unit in the WWTP is further tested. Unlike the simulated benchmark, the measurements characteristics are more complex and they are sampled with different sampling rates in nature. Hence, it is clearer to reveal the efficiency of the proposed model in real applications.

A. Tennessee–Eastman Benchmark

The TE benchmark is designed by the Eastman Chemical Company to provide a simulated industrial process with fundamental control systems [34]. For process monitoring and control purpose, 12 manipulated variables and 41 measured variables are collected which is noted as XMV (1–12) and XMEAS (1–41). In this paper, these variables are collected with three kinds of sampling rates. All the process variables XMEAS (1–22) and manipulated variables XMV (1–12) are sampled per 3 min. For the quality variables, however, the sampling rates are much lower. Among them, XMEAS (23–36) are sampled per 15 min, while XMEAS (37–41) are sampled once an hour.

For model training, XMEAS (1–41) of the normal operation state in 48 h have been collected. Hence, 960 samples of XMEAS (1–22), 192 samples of XMEAS (23–36), and 48 samples of XMEAS (37–41) are used to estimate the model parameters of MR-FA. For comparison, several alternative models are also given. First, the previous work MRPCA is developed using the same multisampling-rate measurements. Besides, the traditional FA is also constructed in different manners. One choice is to use the downsampling method so that a total of 48 samples of XMEAS (1–41) are used to build FA, which is written as FA_1. Another choice is to neglect the sparsely sampled quality samples and 960 samples of XMEAS (1–22) are chosen for FA modeling, which is named as FA_2. The latent variables of MR-FA and MRPCA are chosen as 17 and 10, respectively, using the cross validation. And the factors of FA_1 and FA_2 are given as 8 and 7.

For the fault detection purpose, 21 kinds of traditional faults are tested. The time duration of these faults are also 48 h and all the faults are introduced at the eighth hour. The detailed fault descriptions and fault type can be found in [34]. Given a significance level of 0.99, the missed detection rates of 21 faults using different models are listed in Table I, in which the better results are marked in bold and IDV(0) is the normal test data and the corresponding results are the false alarms. It can be seen that the fault detection results of the proposed MR-FA performs better than the alternatives in most cases, especially for T^2 statistics. For FA_1 using the downsampling method and FA_2 using the fast-sampling measurements, it means that some crucial information and correlation have been abandoned in these models. Hence, it is reasonable that the fault detection performance of FA_1 and FA_2 is deteriorated. It is also noticed that the false alarm of SPE statistics of FA_1 is quite higher than

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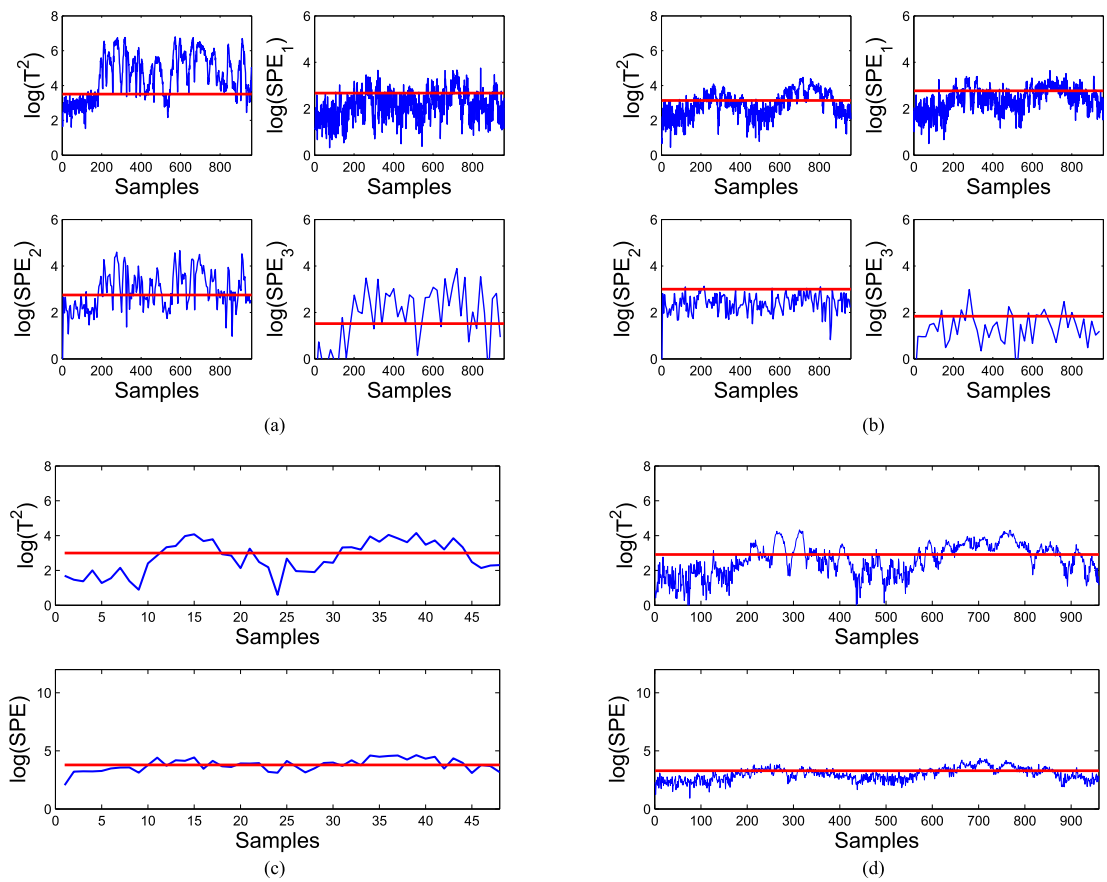

Fig. 3. Fault detection results for IDV(10) in TE process. (a) MR-FA. (b) MRPCA. (c) FA_1. (d) FA_2.

that of other models. Hence, the corresponding missed detection rates are inaccurate. Besides, the assumption of the variance of the process noises are isotropic is too restricted so that the estimations of the post distributions of the latent variables may be affected by such limitations. Hence, it can be found that MR-FA outperforms MRPCA even though both of the models have utilized all the multisampling-rate measurements.

Specially, IDV(10) is taken for instance and the detailed fault detection results are shown in Fig. 3. In IDV(10), a random change is added in the C feed temperature in stream 2, which results in the obvious variations in XMEAS (23, 25, 27, 29, 41). It is noticed that all of these variables are the Moore contents for different components and they are sampled in lower rates compared to the process variables actually. Hence, only using the fast-sampling process variables in FA_2 or a part of downsampling variables in FA_1 will miss some fault-related measurements so that the fault detection performance is much worse in IDV(10) for FA_1 and FA_2. Besides, the noise variations are diverse so that the proposed MR-FA is more proper to depict the cross correlations of the measurements with different sampling rates compared to MRPCA.

B. R2S Anaerobic Reactor Unit

In this subsection, the fault detection results of the proposed MFA models are further tested using a real R2S anaerobic reactor in the papermaking WWTP. The WWTP converts the wastewater into an effluent that can be returned to the water cycle with minimum impact on the environment, or directly reused. A WWTP usually includes blending pond, clarifier, regulating container, anaerobic reactor, cycling standpipe, anoxic pool, biogas storage tank, etc. The detailed flowchart is given in Fig. 4. The papermaking WWTP first transfers impurities into a nonaqueous phase and removes the large suspended solids, sedimentation, and oil slick. Then, the wastewater is

TABLE II
PROCESS VARIABLES DESCRIPTION IN R2S REACTOR

Tags	description
XMEAS(1)	*The inlet flow rate in* R2S-1
XMEAS(2)	*The inlet flow rate in* R2S-2
XMEAS(3)	*The riser level of R2S*
XMEAS(4)	*The water inflow of* R2S-1
XMEAS(5)	*The water circulation rate of* R2S-1
XMEAS(6)	*The inlet flow of PH in* R2S-1
XMEAS(7)	*The outlet flow of PH in* R2S-1
XMEAS(8)	*The water inflow of* R2S-2
XMEAS(9)	*The water circulation rate of* R2S-2
XMEAS(10)	*The inlet flow of PH in* R2S-2
XMEAS(11)	*The outlet flow of PH in* R2S-2
XMEAS(12)	*The inlet* CODcr *value in* R2S-1
XMEAS(13)	*The inlet COD value in* R2S-1
XMEAS(14)	*The inlet COD value in* R2S-2
XMEAS(15)	*The inlet SS value in* R2S-1
XMEAS(16)	*The inlet SS value in* R2S-2
XMEAS(17)	*The inlet PH in* R2S-1 *through the lab analysis*
XMEAS(18)	*The inlet PH in* R2S-2 *through the lab analysis*

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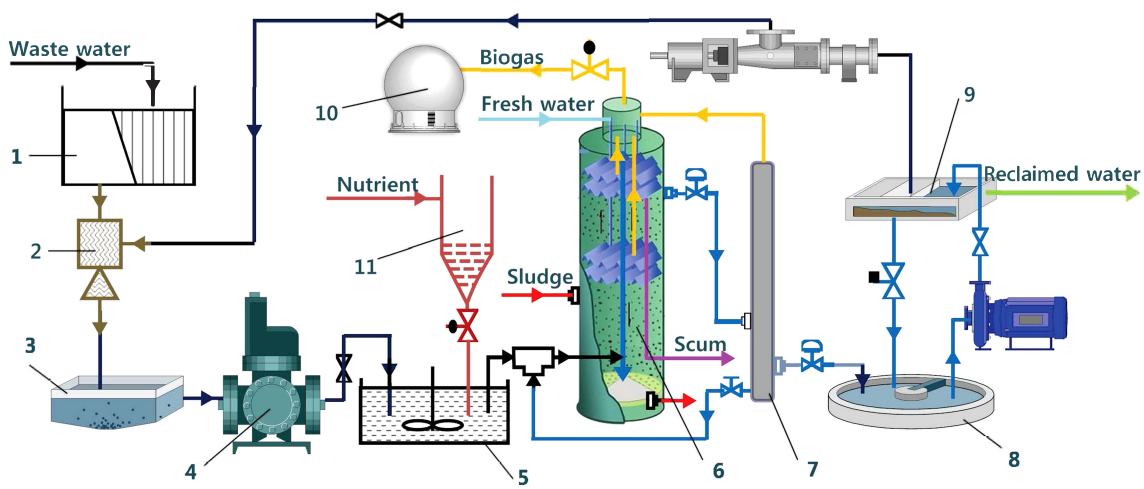

1-Bar screener; 2-Blending pond; 3-Preliminary clarifier; 4-Pump; 5-Regulating container;6-Anaerobic reactor; 7-Cycling standpipe; 8-Anoxic pool; 9-The secondary clarifier; 10-Biogas storage tank; 11-Nutrient auto-count pipette

Fig. 4. Flowchart of the wastewater treatment process.

TABLE III
FALSE ALARM AND MISSED DETECTION RATES IN WASTEWATER TREATMENT PROCESS

Fault	MR-FA				MRPCA				FA_1		FA_2	
	T^2	SPE_1	SPE_2	SPE_3	T^2	SPE_1	SPE_2	SPE_3	T^2	SPE	T^2	SPE
Normal	0.000	0.000	0.008	0.006	0.000	0.000	0.000	0.000	0.038	0.026	0.009	0.003
Fault1	**0.531**	0.924	0.735	**0.912**	0.588	0.925	0.674	1.000	0.559	0.853	0.942	0.565
Fault2	**0.713**	0.914	0.921	1.000	0.748	0.885	0.992	1.000	0.735	0.882	0.914	0.670

biodegraded by microbial anaerobic and aerobic reaction, and finally recycled by secondary sedimentation tank, sand filtration, ultrafiltration, and reverse osmosis. By adding microbial products suitable for improving the water quality, the amount of sludge production and the concentration of chemical oxygen demand will be significantly reduced. Due to the complexity of the operating conditions and the variations in the raw materials, several kinds of faults occur at times.

In this paper, 18 typical process and quality variables in two R2S anaerobic reactors are chosen for process modeling and monitoring purpose. The detailed variable description is listed in Table II. It can be seen that three kinds of sampling rates are included. Among them, XMEAS(1–3) are sampled per hour and XMEAS(4–11) are sampled per 2 h. Both of these two kinds of variables are collected using the online hardware sensors. Also, XMEAS(12–18) are all quality-relevant variables and they are tested at the laboratory and recorded once a day. Using XMEAS(1–18) in 78 days, the proposed MR-FA is constructed. It can be found that 1872 samples of XMEAS(1–3), 936 samples of XMEAS(4–11), and 78 samples of XMEAS(12–18) are collected. Similarly, three alternatives are also built using the same way as the TE process case, which is noted as MRPCA, FA_1, and FA_2. The latent variables of MR-FA and MRPCA are both 3 and the factors of FA_1 and FA_2 are 3 and 2, respectively, using the same model selection method as the TE process case.

For fault detection purpose, two kinds of fault collected from the real operation process are tested. After the proper pretreatment, both of these two faults contains a period of 54 days test data, in which the faults are occurred at 20th day. Given the significance level of 0.99, the false alarm and missed detection rates using different models in the WWTP are listed in Table III, in which normal type means the test normal data and the corresponding results are their false alarms. It can be seen that the final fault detection results are similar to the last case. Especially, the T^2 statistics of MR-FA outperforms several alternatives since it has utilized the sufficient process and quality variables instead of abandoning useful information or destroying the inherent construction in the traditional methods.

V. CONCLUSION

In this paper, a set of MR-FA models are proposed for multirate process modeling and fault detection purpose. Using the complete measurements of different sampling rates, their common factors are determined. The proposed MR-FA is quite different from several independent models, which has been

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discussed adequately both from theoretical and geometric perspective. Also, the multiple fault detection schemes are developed based on MR-FA and its performance has been validated by a simulated TE process and a real R2S anaerobic reactor unit in the WWTP. The results shows that the proposed MR-FA-based methods outperform several alternatives in the multirate process.

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